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Effects of mercury stress on methylmercury production in rice rhizosphere, methylmercury uptake in rice and physiological changes of leaves

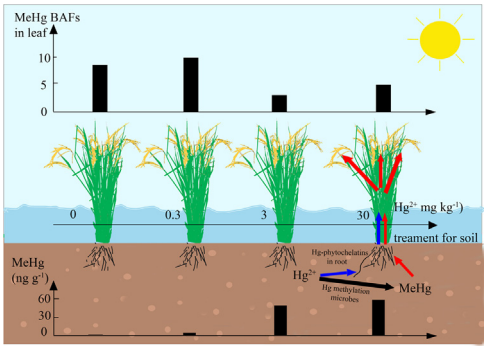
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HIGHLIGHTS

- Highest MeHg/THg occurred with soil mercury concentration of 3 mg kg^{-1}.
- Highest MeHg/THg appeared at the blooming or filling stage.
- MeHg concentrations in seeds reduced to some extent under Hg stress of 30 mg kg^{-1}.
- Little effect on the antioxidant system in leaves with mercury gradient was observed.

GRAPHICAL ABSTRACT



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ABSTRACT

Methylmercury (MeHg) in rice is presumed to be derived from MeHg formed in the soil, although it is still controversial. Moderate soil mercury (Hg) concentration can affect the diversity of soil microorganisms and may also impact the physiological changes and MeHg absorption of rice. In this study, the pot experiment was conducted to explore the effects of Hg concentration gradients (0, 0.3, 3, and 30 mg kg^{-1}) stress on Hg transformation in the rhizosphere, Hg translocation in rice, and physiological changes in rice leaves during the whole rice growing season. Moderate soil Hg concentration (3 mg kg^{-1}) greatly increased the MeHg/THg (1.69%) of rhizosphere, while 30 mg kg^{-1} soil Hg concentration sharply reduced the MeHg/THg (0.29%) of rhizosphere. Highest MeHg/THg of the four groups all appeared at the blooming or filling stage. There was a significant positive correlation between Fe^{2+} in rhizosphere and MeHg/THg, but no significant correlation between SO_4^{2-} and MeHg/THg was observed. Although the 3 mg kg^{-1} soil Hg concentration significantly enhanced MeHg concentrations in seeds, it considerably reduced the bioaccumulation factors of MeHg in roots, stalks, old leaves and young leaves. Soil Hg concentration of 30 mg kg^{-1}, to a certain extent, curtailed MeHg concentrations in seeds, while MeHg concentrations in the husk were significantly increased. Consistent with the result that there was no significant difference for THg concentrations in old and young leaves among the four Hg treatment groups, the content of chlorophyll, H_2O_2, malondialdehyde and antioxidant substances, and the activities of antioxidant enzyme in old and young leaves varied indistinctly among groups.

Main finding: Moderate soil mercury concentration (3 mg kg^{-1}) could extremely enhance MeHg production in the rhizosphere soil and its accumulation in rice; MeHg production in the rhizosphere soil increased greatly at the blooming or filling stage, whereas little effect on antioxidant systems in leaves was observed.

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1. Introduction

Mercury (Hg) is a global pollutant and can migrate long distances, released through natural sources and anthropogenic activities then entered the atmosphere, soil, water and other environmental media (Beckers and Rinklebe, 2017; Obrist et al., 2018). Therefore, Hg will be involved in environmental biogeochemical cycles in various environmental media (Krabbenhoft and Sunderland, 2013). The toxicity of Hg depends largely on its chemical species (Ullrich et al., 2001; Wyn et al., 2009), with methylmercury (MeHg) receiving the greatest attention because it is the most toxic to humans and can be bio-amplified in the food chain (Mcnutt, 2013; Selin, 2018). Although MeHg exposure is mainly related to the consumption of aquatic organisms especially marine fish (Li et al., 2014), rice contaminated with Hg is becoming an emerging health issue in many countries where rice is their staple food (Feng et al., 2008; Rothenberg et al., 2014; Zhang et al., 2010a). Previous studies have found that, the enrichment capacity of rice to MeHg is about 1000 times higher than inorganic Hg (Zhang et al., 2010b) and Hg concentrations in rice growing near Hg-mining areas are far beyond the Chinese National Standard Agency (2017) (THg ≤ 0.02 mg kg^{-1}) (Horvat et al., 2003; Qiu et al., 2006; Zhang et al., 2010b). Since rice is the main cereal crop in the world and the staple food of more than half of the world's population (Peng et al., 2009), an increasing concern has been paid to the exposure of MeHg in rice.

Paddy soil environment is a special wetland ecosystem due to seasonal irrigation. Except for a few co-trophic and fermenting bacteria, the putative *Hg*-methylating microorganisms are strictly anaerobic bacteria or archaea (Gilmour et al., 2013), and the unique anaerobic environment in the paddy field system provides it with excellent condition. Changes in Hg concentrations can greatly affect microbial community structures in soil (Frossard et al., 2017; Ma et al., 2019; Rieder and Frey, 2013). Liu et al. (2014) reported that the moderate Hg concentration (6 mg Hg kg^{-1} DW) increased the diversity of microorganisms, and Frossard et al. (2017) indicated that there were severe diversity loss and shifts in the bacterial and fungal community structures and compositions with Hg concentration of 32 mg Hg kg^{-1} DW. On one hand, the change of Hg concentration affects the microbial community structures of rhizosphere soil (Liu et al., 2014; Ma et al., 2019). On the other hand, since MeHg in rice is highly suspected or proven to be from methylated Hg in the soil (Meng et al., 2011; Zhang et al., 2010b), changes of Hg concentrations in the soil are reputed to have a great impact on the concentration of MeHg in rice. Most studies on Hg accumulation in rice and MeHg production in the rhizosphere soil, however, have specifically focused on the rice maturing stage (Liu et al., 2014; Qiu et al., 2006, 2008; Tang et al., 2017), few studies attached prominence on the characteristics of Hg accumulation in rice throughout the full growing season. Meng et al. (2011) showed that MeHg formed in the rhizosphere was easily absorbed by the roots, mainly stored in stalks and leaves of rice during vegetative growth period, and then transferred to grains during the maturing stage. Strickman and Mitchell (2017) reported that the tillering stage and the flowering stage were the two most important periods for rice to absorb MeHg. During vegetative growth period, rice mainly absorbs nutrients from the soil and the biomass of each tissue gradually increases over time, but during reproductive growth period, rice mainly provides nutrients through leaves. The difference in the rice growing season may not only alter the distribution of Hg in rice tissues, but also shift MeHg production in the rhizosphere soil. Moreover, plants can secrete up to 20% fixed carbon and 15% nitrogen (Haichar et al., 2016; Venturi and Keel, 2016), thus forming a carbon-rich niche in the rhizosphere soil. Considering that the activity of *Hg*-methylating microorganisms can be increased by organic matter (Ullrich et al., 2001), the level of Hg activation in the rhizosphere soil may be different, which will in turn change the degree of Hg methylation in the rhizosphere soil and the enrichment of MeHg in rice.

Furthermore, high levels of Hg^{2+} are highly phytotoxic to plants, causing plants to produce reactive oxygen species (ROS) (Nagajyoti et al., 2010) and inducing visible damage and physiological disorders of plants (Zhang and Tyerman, 1999; Zhou et al., 2007). For example, it triggers the generation of ROS which interferes with the activity of mitochondria and induces oxidative stress, leading to the destruction of cell metabolism of biofilms in plants (Cargnelutti et al., 2006; Messer et al., 2005). In response to the toxicity of heavy metal, plants produce antioxidants (glutathione, ascorbic acid, etc.) and antioxidase (catalase, superoxide dismutase, etc.) to resist oxidative damage (Nagajyoti et al., 2010). Little research, however, has ever been conducted on the effect of Hg stress on chlorophyll, antioxidant substances and antioxidant enzyme activities, and crucially on Hg accumulation and migration in rice during different periods. If changes of Hg concentrations in the soil could induce oxidative damage in rice, it is particularly important to understand how the antioxidant system in rice changes. Therefore, the first objective of this study is to identify one or several growth stages that have a greater impact on Hg uptake in rice and Hg methylation in the rhizosphere soil, and the distribution of MeHg in roots, stalks and leaves at different growth stages. The second objective is to investigate how the enrichment of MeHg in rice responds to the Hg gradient stress and whether there is a risk of Hg exposure under this stress. Finally, the research aims to explore whether oxidative damage in leaves is induced by Hg gradient, and if so, how leaves response to the toxic effect of Hg stress through the antioxidant system is needed to clarify. Results of this study could contribute to a comprehensive understanding about translocation of MeHg from soil into rice plants during the whole rice growing season, and be helpful for the implementation of environmental management strategies to reduce the pollution of MeHg in rice.

2. Material and methods

2.1. Pot experiment and sample collection

The soil used in the experiment was collected from the suburb of Beibei, Chongqing, China, and its average THg concentration was 0.15 mg kg^{-1}, which was close to the average background level of THg in Chinese agricultural soil (0.10 mg kg^{-1}) (Chinese National Standard Agency, 2017). The soil was air-dried and mashed, then sieved with 20 mesh for rice pot experiment. Each plastic pot (90 cm (length) × 36 cm (width) × 25 cm (height)) was filled with 40 kg sieved soil. 10 g l^{-1} Hg^{2+} solution was added to the pot, resulting in soil Hg concentrations of 0 mg kg^{-1}, 0.3 mg kg^{-1}, 3 mg kg^{-1} and 30 mg kg^{-1} marked as Z (Zero), M (Middle), H (High) and V (Very High) groups respectively. The Hg concentration gradient referred to the XiuShan YangShikeng Hg mining area in southwestern China (Ma et al., 2019; Xu et al., 2017), and similar Hg concentrations appeared in other studies (Frossard et al., 2017; Harris-Hellal et al., 2009; Philippot et al., 2008; Rieder and Frey, 2013). Base fertilizers ($CO(NH_2)_2$, 150 mg kg^{-1}; Ca $(H_2PO_4)_2$, 100 mg kg^{-1}; KCl, 85 mg kg^{-1}) were added to the pot. After the mixture was stirred well, the water surface was kept 3–5 cm away from the soil surface during standing about two weeks and then transplanted the rice seedlings in the pot in greenhouse. During the cultivation in the greenhouse, water surface above the soil in pot was still maintained at 3 to 5 cm. The space of the plastic pots was large enough to allow three replicates in a plastic pot. Considering the four Hg concentrations and five sampling time points (reviving, tillering, blooming, filling and maturing stages), a total of 20 pots and 60 rice plants were prepared. The reviving, tillering, blooming, filling and maturing stages were respectively corresponded to 20, 40, 80, 105, 122 days after rice seedlings were transplanted into the plastic pots on May 3, 2019. The collection method of rhizosphere soil was the same as that of Ma et al. (2019). The whole rice plant was dug out of the soil, and the large blocks not containing roots were gently shaken off. Then the soil attached to the surface of the rice roots was collected. Rice in plastic bags with their respective labels were transported back to the laboratory to collect

the underground (roots) and aboveground (stalks, old leaves, young leaves, husks and seeds) parts. Wash the roots of rice with tap water to remove debris, and finally rinse with deionized water to achieve the purpose of removing root soil. Samples of leaves were made in duplicate, one was stored directly in the −20 °C refrigerator for determination of biochemical indexes, and the remaining one, roots, old leaves, young leaves, husks and seeds were stored in the −20 °C refrigerator for determination of Hg after freeze-dried.

2.2. Determination of physicochemical property in rhizosphere soil and Hg concentrations in rhizosphere soil and rice

After SO_4^{2-} in soil samples was extracted by water, the extraction was analyzed by ion chromatograph. Using MultiC/N2000 carbon and nitrogen automatic analyzer to determinate organic matter (OM) directly. Connecting a pH meter (FEP20-FiveEasy Plus, Mettler-Toledo GmbH, Switzerland) with a glass electrode to measure pH values of each soil sample (soil:extractant = 1:2). Fe^{2+} content in soil samples was measured by Optima 3000 (Perkin-Elmer) induction plasma atomic analysis. The determination methods of THg and MeHg in rice tissues and rhizosphere soils were the same as those of Ma et al. (2017) and Tang et al. (2017). Rhizosphere soil and rice samples were digested by 1:3 HCl + HNO_3 and 4:1 HNO_3 + H_2SO_4 respectively, and then tested by GC-CVAFS to determine THg concentrations. The samples of rice and rhizosphere soil were digested with KOH methanol/solvent extraction and $CuSO_4$-methanol/solvent extraction, respectively, and then GC-CVAFS detection was performed to determine MeHg concentrations.

2.3. Determination of oxidation and anti-oxidation indexes in leaves

Oxidative damage to plant cells after Hg stress has been shown, due to the excessive accumulation of reactive oxygen species (ROS), including free radicals and H_2O_2. Although superoxide ($O\bullet_2^-$) or H_2O_2, produced through the oxygen reduction by many intracellular reactions, are not very reactive, they can form hydroxyl radicals (-OH) (Cadenas, 1989; Halliwell and Cutteridge, 1990; Nagajyoti et al., 2010) and excessively high concentrations of H_2O_2 combined with ROS can cause lipid peroxidation by attacking membrane lipids (Janero, 1900), which may be the reason of most oxidative damage in biological systems. Proline, AsA, and GSH in leaves play an important role in detoxification of reactive oxygen species in plants and their accumulation was also observed in various plants under heavy metal stress (Liu et al., 2011; Prasad and Saradhi, 1995; Shah and Dubey, 1998). The levels of H_2O_2, chlorophyll and MDA could mirror the degree of oxidative damage in plants under heavy metal stress, while the antioxidant AsA, GSH and proline, as well as, antioxidant enzymes POD, CAT and SOD can resist the toxic effects of heavy metals on plants (Nagajyoti et al., 2010; Zhang and Tyerman, 1999; Zhou et al., 2007). The analysis methods of these indexes in leaves were as follows. After 100 mg of leaf powder grounded in liquid nitrogen was extracted with 96% (v/v) ethanol in a hot water bath at 60–80 °C until green leaves faded, the absorbance of the extraction detected at 645 and 663 nm by spectrophotometry was used to calculate the chlorophyll content (Lu et al., 2017). 0.1 g of leaves was homogenized with a mortar and pestle in 1.5 ml phosphate buffer (50 mM, pH 6.5), containing 1 mM hydroxylamine. After centrifugation at 10,000g for 10 min, 0.5 ml supernatant was mixed with 1.5 ml 20% (v/v) H_2SO_4, containing 0.1% (v/v) $TiCl_4$. After the mixture was centrifuged at 10,000g for 10 min, the absorbance of the supernatant detected at 410 nm by spectrophotometry was used to calculate the H_2O_2 content (Chen et al., 2009). By homogenizing 1 g of leaves (fresh weight) in 10 ml of cold 5% (w/v) metaphosphoric acid, centrifuging the homogenized liquid (22,000g, 4 °C) for 15 min, collecting the supernatant for analyzing Ascorbate (AsA) and glutathione (GSH) (Chen et al., 2009). The Proline content in leaves was determined by the acidic ninhydrin method (Sun et al., 2006). The content of malondialdehyde (MDA) in leaves was determined by the thiobarbituric acid method (Janero, 1900). The guaiacol peroxidase (POD) activity was determined using the guaiacol color method (Amako et al., 1994). Catalase (CAT) activity was measured using ultraviolet absorption method (Hernandez et al., 2010). The activity of superoxide dismutase (SOD) was determined by measuring the inhibition of the photochemical reduction of nitroblue tetrazole (NBT) (Beauchamp and Fridovich, 1971).

2.4. Quality control and statistical analysis

The detection limits of THg and MeHg in rice were 0.01 μg kg^{-1} and 0.002 μg kg^{-1}, respectively, while in rhizosphere soil were 0.2 μg kg^{-1} and 0.01 μg kg^{-1}, respectively. In the determination of THg, GSS-7 (GBW07407, Institute of Geophysical and Chemical Exploration) was used as the standard substance, with calibration value of 61 ± 6 ng g^{-1} and the recovery rate of 90–120%. The SPSS 21.0 was used for statistical analysis and Origin 2019 was utilized for drawing pictures. One-way analysis of variance (ANOVA) was used to test significant differences among treatments. The correlation coefficient (r) and significant probability (p) of linear regression fitting were calculated and p value of less than 0.05 was considered to indicate statistical significance.

3. Results and discussion

3.1. THg and MeHg in rhizosphere soil

The dynamic changes of THg in rhizosphere soil with the growing season of rice were shown in Fig. 1A. The THg concentration in the rhizosphere soil in group Z gradually decreased with time. As for group M, THg concentrations (0.41 ± 0.004 mg kg^{-1}) in the rhizosphere soil at the tillering stage was minimized among the rice growing season in group M. The THg concentration in the rhizosphere in group H decreased to 2.65 ± 0.22 mg kg^{-1} at the tillering stage, but there was no significant difference for THg concentrations among the rice growing season ($p > 0.05$). The THg concentration in the rhizosphere soil in group V decreased to a minimum of 14.56 ± 0.04 mg kg^{-1} at the blooming stage, and reached a maximum of 37.18 ± 6.88 mg kg^{-1} at the filling stage. Phytochelate peptides that have a detoxifying effect on Hg in rice can effectively chelate Hg^{2+} in rice to form Hg-PCs compounds promoting the transport of Hg^{2+} from the cytoplasm to vacuoles and retaining Hg^{2+} in the roots (Krupp et al., 2009). Besides, the inorganic Hg in rice mainly enters the body through the free diffusion mechanism of the root system to absorb water or inorganic salts (Meng et al., 2014). Although rice absorbs Hg in the soil, the growth of roots and the increase in root biomass from reviving stage to blooming stage may contribute to the decrease of THg concentrations in roots during the tillering or blooming stages. When the concentration of inorganic Hg fixed in the root is more than the concentration of inorganic Hg in the rhizosphere soil, this may affect the absorption of inorganic Hg in rice, resulting in a certain increase of THg concentrations in rhizosphere soil after the blooming stage. However, the change of THg concentrations in rhizosphere soil in group H was not affected by the rice growth stage, which may indicate that the treatment of 3 mg kg^{-1} Hg concentration affected the formation of phytochelate peptides in roots.

MeHg concentrations in rhizosphere soils of group Z changed slightly (Fig. 1B), with no significant difference among different growing seasons ($p > 0.05$). MeHg concentrations in the rhizosphere soil of group M increased first and then decreased, reaching the maximum of 11.00 ± 1.65 ng g^{-1} at the blooming stage. MeHg concentrations in rhizosphere soils of group H and V reached maximum at the filling stage, which were 85.38 ± 0.04 ng g^{-1} and 73.59 ± 14.19 ng g^{-1} respectively. Moreover, MeHg concentrations in group H and V differed greatly at the blooming stage (H, 27.77 ± 1.65 ng g^{-1}; V, 71.67 ± 6.67 ng g^{-1}), and the other rice growing season were almost the same (Fig. 1B). The ratios of MeHg to THg concentration THg (MeHg/THg) in rhizosphere soils varied with the rice growing season, presenting almost the same trend of MeHg concentrations in rhizosphere soils (Fig. 1C). In the four Hg

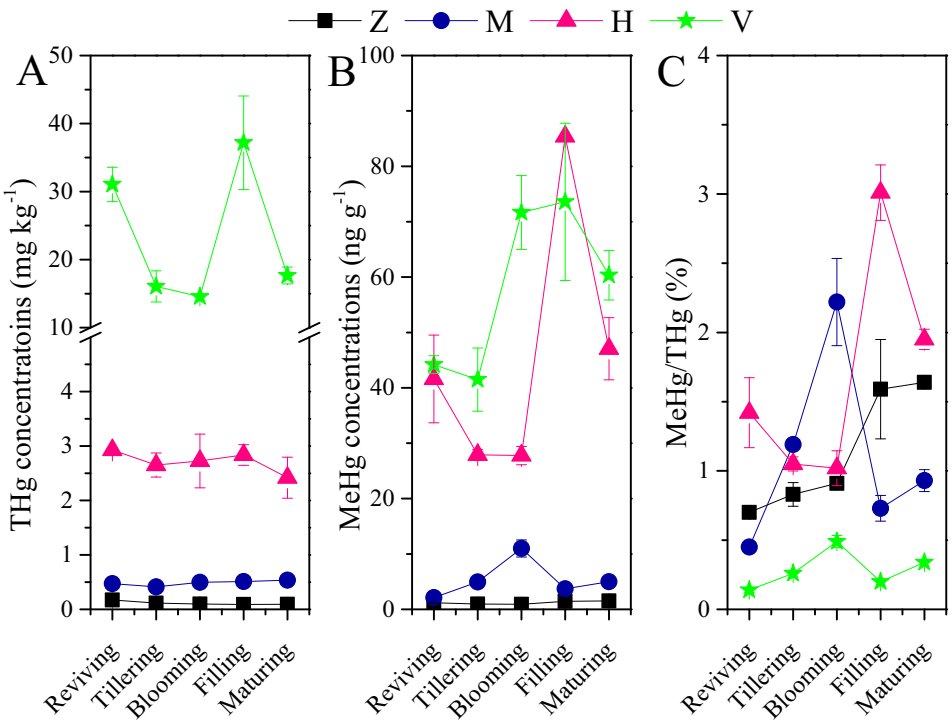


Fig. 1. Dynamic changes of THg, MeHg and MeHg/THg in the rhizosphere soil (Z, M, H and V represent 0, 0.3, 3, and 30 mg kg^{-1} soil Hg concentrations, respectively.)

treatment groups, the maximum value of MeHg/THg all appeared in the blooming stage or filling stage. The MeHg/THg ratios in the four Hg treatment groups were in descending order of group H, M, Z and V, with average values of 1.69%, 1.10%, 1.13%, and 0.29%, respectively. The yield of MeHg and the degree of Hg methylation in soil can be measured by the ratios of MeHg to THg concentration (MeHg/THg) in soil (Kainz and Lucotte, 2006). Generally, the concentration of MeHg accounts for about 0.1% of the THg concentration in soils and sediments, reaching a maximum of 1.5% (Avramescu et al., 2011; Ullrich et al., 2001). In this study, probably due to Hg^{2+} as the Hg species added into the soil, MeHg/THg in rhizosphere soils was mostly more than 1.0%, and some even reaching 2.0%, implying that paddy soil was a hot spot for Hg methylation. The Hg^{2+} concentration of 3 mg kg^{-1} promoted the methylation of Hg in the soil, especially during the filling stage (3.01%). In the four groups, the bioavailable Hg, especially water-soluble Hg (WHg), would enhance with the increased Hg^{2+} concentration in soil, which strongly affected the microbial community structure in forest soil (Rieder and Frey, 2013; Frossard et al., 2017), and contributed to the increase of Hg-resistant bacteria in Hg-polluted paddy soil (Hu et al., 2019). With the WHg enhancement, Hg-sensitive bacteria would be killed or disabled, whereas Hg-resistant bacteria would be promoted (Hu et al., 2019). Connell (1979) reported that intermediate disturbance could result in high biodiversity. The 3 mg kg^{-1} Hg concentration in this study could affect the activity or diversity of soil Hg methylating microorganisms because of the increased microbial diversity under moderate Hg stress (6 mg Hg kg^{-1} DW) (Liu et al., 2014). However, Frossard et al. (2017) found that the highest Hg concentration (32 mg Hg kg^{-1} DW) sharply shifted the microbial community structure and compositions. Therefore, the Hg methylation process of soil microbial may be greatly inhibited under 30 mg kg^{-1} Hg stress, resulting in MeHg/THg in rhizosphere soil in group V being less than that in the control group Z (Fig. 1C).

In addition to the effect of changes in Hg concentrations, the Hg methylation in the rhizosphere soil was obviously promoted at the blooming or filling stages. An increased pH in the rhizosphere of the four groups during the rice growing season was observed (Fig. S1A). As mentioned earlier, plants can secrete up to 20% fixed carbon and 15% nitrogen (Haichar et al., 2016; Venturi and Keel, 2016), thus forming a carbon-rich niche in the rhizosphere soil. Therefore, it may be related to the secretions in the rhizosphere released by rice during vegetative growth period when pH in the rhizosphere at this period was lower than 7. OM in rhizosphere soils in the four groups decreased gradually with the rice growing season, while that at vegetative growth period was higher than that at reproductive growth period (Fig. S1B). SO_4^{2-} at vegetative growth period was obviously higher than that at reproductive growth period, with the decrease of SO_4^{2-} occurring at the blooming stage (Fig. S1C). Fe^{2+} had no remarkable changes between vegetative and reproductive growth periods, but the enhanced Fe^{2+} appeared at the blooming stages (Fig. S1D). The results of these physicochemical properties were consistent with the changes of MeHg concentrations and MeHg/THg in rhizosphere soils. Low pH and anaerobic conditions will not only increase methylation rates, but also reduce demethylation rates, resulting in a net production of MeHg (Gilmour et al., 1992). Dissolved organic matter can serve as a substrate for microorganisms and stimulate Hg methylation (Gerbig et al., 2012). Methylated bacteria are strictly anaerobic bacteria, including sulfate reducing bacteria (SRB) and iron reducing bacteria (FeRB) using sulfate, iron and carbon dioxide as terminal electron acceptors (Gilmour et al., 2013). These physicochemical properties are closely related to the process of Hg methylation. However, through the correlation analysis between these physicochemical properties and MeHg/THg (Table S1), only ferrous ions had a significant positive correlation with MeHg/THg ($r = 0.27$; $p < 0.05$), which may indicate that FeRB consuming Fe^{3+} for Hg methylation process may be more active during the flowering stage. Therefore, the highest Hg methylation efficiency occurred at the flowering or the filling stages. MeHg production in the environment is closely related to microorganisms including Deltaproteobacteria, Firmicutes, and Methanomicrobia (Gilmour et al., 2013). The increased Hg^{2+} concentrations could lead to the inhibition and death of Hg-sensitive bacteria, and the increase of Hg-resistant microorganisms (Hu et al., 2019). It may be the reason why there is no significant correlation between organic matter and MeHg/THg in the rhizosphere. MeHg was thought to be produced predominantly by SRB in anoxic sediments, but some studies found that the methylation rate of FeRB could rival

that of SRB (Fleming et al., 2006; Kerin et al., 2006) and compared with FeRB and SRB, the methylation rate of methanogens was relatively low (Gilmour et al., 2013). Liu et al. (2018) found that there was a higher abundance of FeRB and methanogens, and less abundance of SRB (<3%) when methylators in paddy soils were grouped into the three typical functional clades. It may suggest that FeRB play a crucial role in contributing to MeHg production in paddy soil, which needs to be further confirmed in the future.

3.2. MeHg in rice tissues

The dynamic changes of MeHg concentrations in rice tissues during the rice growing season were shown in Fig. 2. In general, the concentration of MeHg in rice tissues showed a gradual decrease from the reviving stage to the maturing stage. MeHg concentrations in stalks and leaves were highest at the tillering stage, being consistent with the result of Strickman and Mitchell (2017) that maximum absorption of MeHg occurred before blooming stage. However, some research reported that when MeHg concentrations was multiplied by the rice biomass to obtain the total MeHg mass of each tissue part of rice, MeHg mass in rice tissue was the largest during the flowering stage (Meng et al., 2011; Strickman and Mitchell, 2017), suggesting that tillering and blooming stages are the two most important periods for rice to absorb MeHg. MeHg concentrations in roots, stalks and leaves were the lowest at the filling and the maturing stage (Fig. 2). The MeHg in soil is first absorbed by the rice roots which is likely to bind to the protein and be transported to the aerial parts (mainly stalks and leaves) during vegetative growth period, and it is transported together with the protein and enriched in seeds during the maturing stage (Meng et al., 2011, 2014), leading to a decrease in the MeHg concentration in roots, stalks, and leaves during the filling and maturing stage. Furthermore, the growth dilution before reproductive period of plants may reduce MeHg concentrations in the plants (Meng et al., 2011; Strickman and Mitchell, 2017). Thus, MeHg concentrations in stalks and leaves were the highest at the tillering stage, while MeHg concentrations in roots, stalks and leaves were the smallest at the filling and the mature stage. The MeHg concentration in leaves in group V was significantly different from that in group Z ($p < 0.001$), M ($p < 0.001$) and H ($p < 0.001$), indicating that MeHg concentrations in old and young leaves in group V was significantly higher than that in other groups (Fig. 2C & 2D). The MeHg concentration in husks in group V (1.11 ± 0.19 mg kg^{-1}) was significantly higher than that in other Hg treatment groups, while MeHg concentrations in seeds in group H (3.74 ± 0.63 mg kg^{-1}) were significantly higher than that in other Hg treatment groups. Compared with seeds in group H, MeHg concentrations (1.95 ± 0.48 mg kg^{-1}) in seeds in group V decreased to some extent (Fig. 3), showing that MeHg concentrations in seeds was significantly increased under Hg stress of 3 mg kg^{-1} and MeHg concentrations in husks significantly increased under Hg stress of 30 mg kg^{-1}. The concentration of MeHg in rhizosphere soils had a significant positive correlation with that in rice tissues, while having an insignificant correlation with that in seeds (Table S3). In addition, no significant correlation for MeHg concentrations between seeds and husks was found (Table S3). Results of this research seemed to be inconsistent with other related studies (Meng et al., 2014; Tang et al., 2017; Zhang et al., 2010b) that there was a significant positive correlation for MeHg concentrations between seeds and soils, or seeds and husks. However, when MeHg concentration data of seeds and husks in group V were removed, MeHg concentrations in seeds were highly positively correlated with that in rhizosphere soils ($r = 0.92$, $p < 0.01$), and in husks ($r = 0.99$, $p < 0.05$) (Table S4), potentially indicating that soil MeHg was the source of MeHg in rice. As discussed in Section 3.1, Hg^{2+} concentration of 3 mg kg^{-1} may affect the process of MeHg production by impacting the microbial community structure in paddy soil. The MeHg in rice is thought to come from the MeHg formed in soil (Tang et al., 2017; Zhang et al., 2010b). The highest MeHg concentration in the rhizosphere was observed in H and V groups (Fig. 1B),

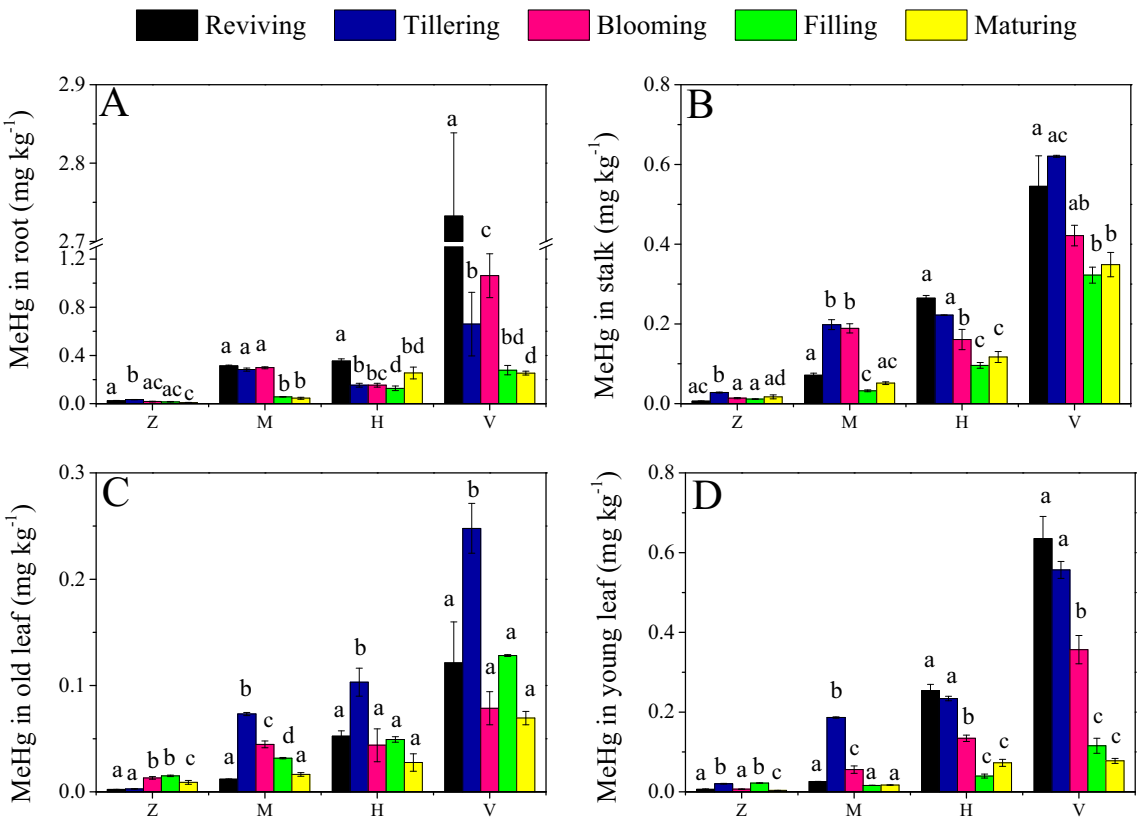


Fig. 2. Dynamic changes of MeHg concentrations in roots, stalks, old and young leaves (Z, M, H and V represent 0, 0.3, 3, and 30 mg kg^{-1} soil Hg concentrations, respectively. Different letters at the top of the bar graph represent significant differences among the rice growing season.)

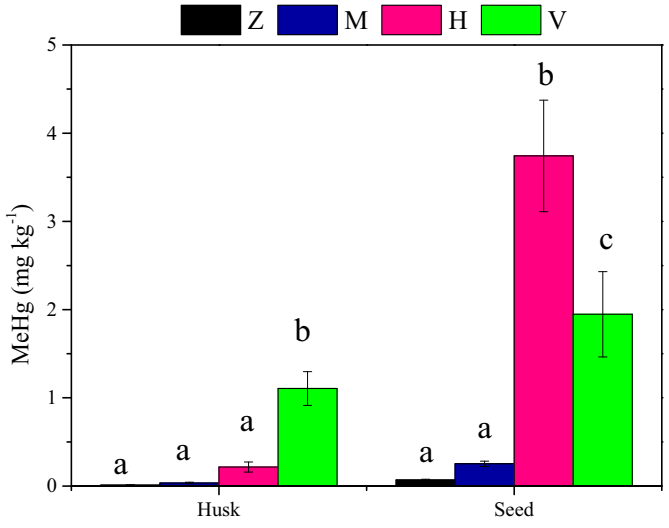


Fig. 3. MeHg concentrations in husks and seeds under different Hg concentration treatments (Z, M, H and V represent 0, 0.3, 3, and 30 mg kg^{-1} soil Hg concentrations, respectively. Different letters at the top of the bar graph represent significant differences among Z, M, H and V group.)

thus MeHg concentrations in seeds of H and V groups were significantly higher than that of other groups (Fig. 3). When the concentration of soil Hg or the accumulation of MeHg in rice exceeds a certain limit, rice may preferentially transfer MeHg into husks, thereby slowing down the accumulation of MeHg in seeds. This can be reflected by the results that MeHg concentrations in seeds were 17.36 and 1.76 times of that in husks of group H and V, respectively.

When the bioaccumulation factors surpass 1, the bioaccumulation is strong (Patra and Sharma, 2000). As a result, rice has a strong MeHg enrichment ability, and seeds of rice have the strongest MeHg enrichment ability compared with other rice tissues (Table S2). Although the MeHg bioaccumulation factors of seeds in group H maximized, MeHg bioaccumulation factors of roots, stalks and leaves in group H were the lowest compared to other Hg treatment groups. Additionally, it was interesting to find that MeHg concentrations in roots, stalks and leaves in group Z, M and H were similar (Fig. 2), whereas MeHg concentrations in rhizosphere soils in group Z, M, and H were quite different (Fig. 1B). *In planta* demethylation, observed in live rice plants (Li et al., 2016), occurs via endophytic microorganisms or abiotic photolysis (Lin et al., 2012) and may further reduce Hg^{2+} to Hg^0 (Battke et al., 2005), indicating that there might be a process of demethylation in rice, resulting in the lowest MeHg bioaccumulation factors of roots, stalks and leaves in group H.

3.3. Rice leaves response different Hg concentration stress

3.3.1. THg in rice leaves

The changes of THg concentrations in old and young leaves during the rice growing season were shown in Fig. 4A. Generally, probably due to the increased biomass of rice during vegetative growth period, THg concentrations in old and young leaves in the four Hg treatment groups at tillering, blooming and filling stages varied greatly compared with the other rice growing period. However, there was no significant difference for THg concentrations between old and young leaves in group H among the rice growing season. These findings were in line with the result that the change of THg concentrations in rhizosphere soils in group H was not affected by the rice growing period. THg bioaccumulation factors of old and young leaves in group H (old leaf, 0.08; young leaf, 0.02) and V (old leaf, 0.06; young leaf, 0.02) were lower than that in group Z and M (Table S5), suggesting that Hg concentration of 3 mg kg^{-1} might impact the detoxification mechanism of rice to Hg. Inorganic Hg (IHg) in rice leaves may be absorbed from soil (Zhang et al., 2010b), the atmosphere (Meng et al., 2010), or a mixture of these sources (Strickman and Mitchell, 2017). Xu et al. (2019) found that after Se supplementation, the apparent reduction of IHg in rice tissues was observed except for rice leaves, potentially implying that IHg in rice leaves is primarily from the atmosphere (Fay and Gustin, 2007). In this study, however, THg concentrations in old and young leaves among the four Hg treatment groups were $V > H > M \approx Z$ and $V > H \approx M > Z$ respectively, and the THg concentration in leaves in group V was significantly different from that in group Z ($p < 0.001$), M ($p < 0.001$) and H ($p < 0.001$), indicating that the THg concentration in leaves in group V was significantly higher than that in other groups. Therefore, combined with the same atmospheric Hg, most of Hg in leaves was transported from soil at least with Hg^{2+} treatments.

3.3.2. Chlorophyll, MDA and H_2O_2 in rice leaves

The chlorophyll and H_2O_2 content in old and young leaves were not significantly different among the four Hg treatment groups (Fig. 4B & D). MDA, as an indicator of lipid peroxidation levels, in old and young leaves was also differed insignificantly among the four groups (Fig. 4C). Results here accorded with that there was no significant difference for THg concentrations in old leaves among the four groups, probably revealing that the Hg gradient in this study had little effect in inducing visible damage and physiological disorders of rice leaves. Cargnelutti et al. (2006) showed that the levels of lipid peroxides in cucumber seedlings increased at the highest level of Hg (500 μM $HgCl_2$). Moreover, some studies found that the biomass of roots and shoots was reduced under Hg stress (Cargnelutti et al., 2006; Zhou et al., 2007). Liu et al. (2011) reported that high concentration of Cd resulted in the death of *T. patula* root cells, thereby reducing MDA contents. A significant negative correlation occurred between THg concentrations and MDA contents in old leaves ($r = -0.325$, $p < 0.05$), indicating that it might be related to the senescence and death of cells in old leaves. In contrast, a significant positive correlation appeared between THg concentrations and H_2O_2 contents in young leaves ($r = 0.399$, $p < 0.05$), consisting with the significant difference for THg concentrations in young leaves between group Z and H, and suggesting that Hg concentration of 3 mg kg^{-1}, to some extent, induced oxidative damage to young leaves.

3.3.3. Antioxidants in leaves

The content of proline, AsA and GSH in old leaves at the five growth stages had no significant difference among the four Hg treatment groups (Fig. 5A, B & C). The content of proline in young leaves at the reviving stage in group H (80.18 ± 2.76 μg g^{-1} FW) was significantly lower than that in group Z (229.68 ± 17.59 μg g^{-1} FW), while the content of proline in young leaves at the blooming stage in group H (300.40 ± 18.59 μg g^{-1} FW) was significantly higher than that in group Z (155.62.18 ± 51.46 μg g^{-1} FW). At the blooming stage, AsA contents in young leaves in group H (4.00 ± 0.79 μmol g^{-1} FW) was significantly lower than that in group Z (7.79 ± 0.07 μmol g^{-1} FW). No significant difference for the content of GSH in young leaves at five rice growth stages appeared among the four Hg treatment groups (Fig. 5C). In this study, the accumulation of proline in young leaves in group H occurred, but it was not observed that the accumulation of AsA and GSH in old and young leaves, which might reveal that rice with Hg concentration of 3 mg kg^{-1} reduced the toxic effects of Hg to itself by the accumulation of proline in young leaves. However, the correlation analysis between THg concentrations and antioxidant substances contents in young leaves revealed that there was no significant correlation between them (Table S6), suggesting that the Hg gradient in this study was limit for causing oxidative damage in leaves.

3.3.4. Antioxidant enzymes activities in leaves

In response to heavy metal stress, the activity of antioxidant enzymes (such as SOD and POD) in plants increased (Prasad et al., 1999; Verma and Dubey, 2003), while, in this study, the contents of POD,

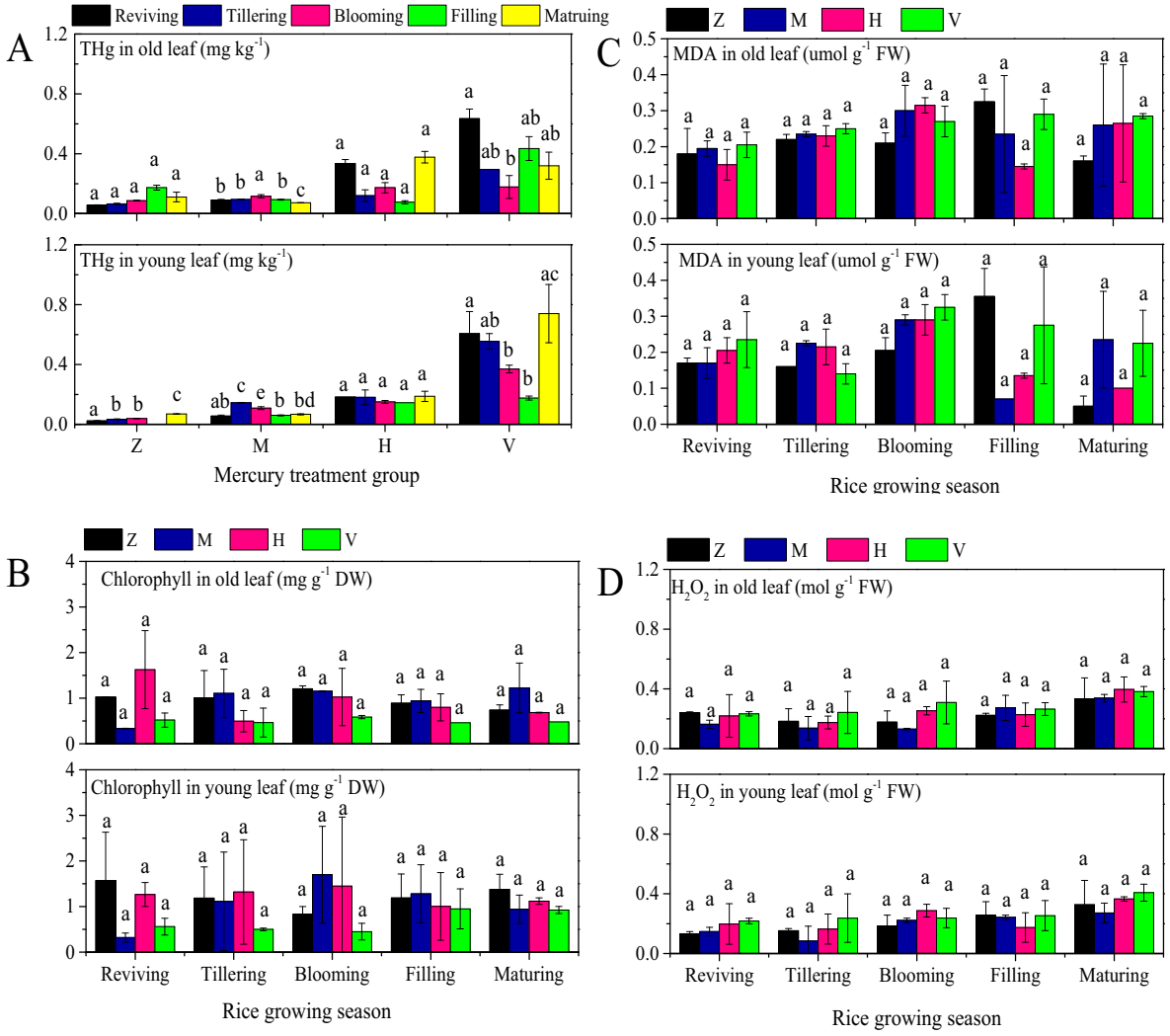


Fig. 4. THg, chlorophyll, MDA and H_2O_2 in old leaves and young leaves under different Hg concentration treatments (Z, M, H and V represent 0, 0.3, 3, and 30 mg kg^{-1} soil Hg concentrations, respectively. Values with the same letter are not significantly different at $p > 0.05$).

SOD and CAT in old and young leaves at the five growth stages were insignificantly different among the four groups (Fig. 5D, E & F), implying that the Hg gradient stress in this study had little effect on antioxidant enzyme activities in leaves. Indeed, it was observed that the accumulation of AsA and GSH and the enhancement of antioxidant enzyme activities in rice leaves under Hg stress occurred (Chen et al., 2009). It may be because the mean of THg concentrations in old and young leaves in this study was much smaller than those in the shoots of *P. vittata* and *N. exaltata* in the study of Chen et al. (2009) (Table S7), indicating that the Hg gradient in this study had little effect on the antioxidant system in rice leaves. Inorganic Hg mainly enters rice roots through free diffusion mechanism (Meng et al., 2014) and the chelate peptides in rice effectively chelate Hg^{2+} to form Hg-PCs compounds retaining Hg^{2+} in roots (Krupp et al., 2009). Because these chelate peptides cannot chelate MeHg, they have little effect on the absorption of MeHg for rice. Moreover, while rice absorbed nutrients, MeHg was mistakenly used as a nutrient element or MeHg would be absorbed into rice along with certain nutrient elements (Tang et al., 2017). Consequently, most of the Hg^{2+} was retained in the root of rice, and MeHg was easily transferred from the soil to the aboveground part of rice. Besides, MeHg/THg of old and young leaves were greater than those of roots, and the higher the Hg concentration was, the more obvious it was (Table S8). Therefore, possible reasons were advanced for its little effect on the antioxidant system in leaves under the Hg gradient stress. Firstly, it might be the role of rice roots resulting in changes of Hg concentrations had little effect on antioxidant systems in leaves. Secondly, it was possible that a large part of THg in leaves was MeHg or the Hg^{2+} content in leaves was not high, which had little effect on the antioxidant system in leaves. Finally, the possible demethylation process and the reduction of Hg^{2+} to Hg^0 in rice affects the response of antioxidant system of leaves to the Hg gradient stress.

4. Conclusion

MeHg production in the rhizosphere partly enhanced at the rice growing season, especially the blooming and filling stages. The treatment of Hg concentration of 3 mg kg^{-1} could greatly increase the Hg methylation efficiencies of rhizosphere soils, especially during the filling period (3.01%), which may be related to the activity of iron-reducing bacteria in rhizosphere soils and enhanced bioavailabilities of Hg. This Hg concentration significantly increased the concentration of MeHg in rice seeds. The treatment of Hg concentration of 30 mg kg^{-1} greatly reduced the MeHg/THg of the rhizosphere soil (0.29%) and reduced the MeHg concentration to some extent, but significantly increased the MeHg concentration in husks, potentially implying that rice may slow down the enrichment of MeHg in seeds by transporting MeHg in rice into husks. Probably due to the demethylation in rice, the lowest MeHg bioaccumulation factors of roots, stalks, old and young leaves were observed under Hg stress of 3 mg kg^{-1}. Thus, the treatment of Hg concentration of

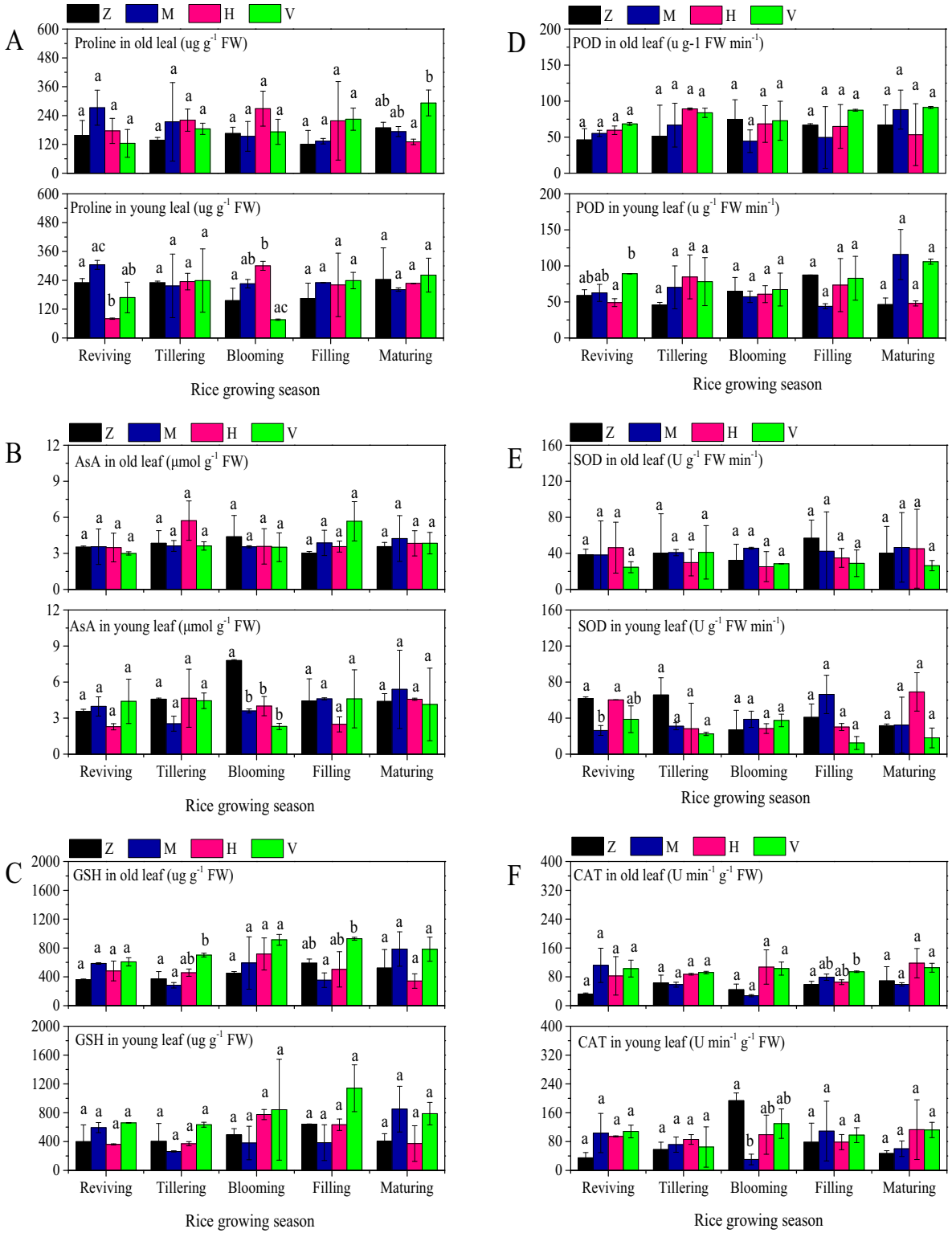


Fig. 5. Proline, AsA, GSH, POD, SOD and CAT in old leaves and young leaves under different Hg concentration treatments (Z, M, H and V represent 0, 0.3, 3, and 30 mg kg^{-1} soil Hg concentrations, respectively. Values with the same letter are not significantly different at $p > 0.05$).

3 mg kg^{-1} extremely affected the transformation of Hg in rhizosphere soil and the translocation of MeHg in rice compared with the other Hg concentration. However, possibly because of the role of roots, the concentration of Hg^{2+} in leaves, and the demethylation in rice, the Hg gradient in this study had little effect on the content of oxidizing and antioxidant substances, and the activity of antioxidant enzymes in leaves. Our results may be beneficial to the implementation of effective engineering or environmental management strategies about reducing MeHg pollution in rice.

CRediT authorship contribution statement

Pan Guo: Writing - original draft, Software, Visualization, Data curation, Writing - review & editing. **Hongxia Du:** Visualization,

Investigation, Supervision, Writing - review & editing. **Dingyong Wang:** Supervision, Validation. **Ming Ma:** Conceptualization, Methodology, Supervision, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2020.142682.

References

Amako, K., Chen, G.X., Asada, K., 1994. Separate assays specific for ascorbate peroxidase and guaiacol peroxidase and for the chloroplastic and cytosolic isozymes of ascorbate peroxidase in plants. Plant Cell Physiol. 3 (3), 497–504. https://doi.org/10.4028/www.scientific.net/MSF.492-493.229.

Avramescu, M.L., Yumvihoze, E., Hintelmann, H., Ridal, J., Fortin, D., Lean, D.R.S., 2011. Biogeochemical factors influencing net mercury methylation in contaminated freshwater sediments from the St. Lawrence River in Cornwall, Ontario, Canada. Sci. Total Environ. 409, 968–978. https://doi.org/10.1016/j.scitotenv.2010.11.016.

Battke, F., Ernst, D., Halbach, S., 2005. Ascorbate promotes emission of mercury vapour from plants. Plant Cell Environ. 28 (12), 1487–1495. https://doi.org/10.1111/j.1365-3040.2005.01385.x.

Beauchamp, C., Fridovich, I., 1971. Superoxide dismutase: improved assays and an assay applicable to acrylamide gels. Anal. Biochem. 44, 276–287. https://doi.org/10.1016/0003-2697(71)90370-8.

Beckers, F., Rinklebe, J., 2017. Cycling of mercury in the environment: sources, fate, and human health implications: a review. Crit. Rev. Environ. Sci. Technol. 47 (9), 693–794. https://doi.org/10.1080/10643389.2017.1326277.

Cadenas, E., 1989. Biochemistry of oxygen toxicity. Annu. Rev. Biochem. 58, 79–110. https://doi.org/10.1146/annurev.bi.58.070189.000455.

Cargnelutti, D., Tabaldi, L.A., Spanevello, R.M., Jucoski, G.O., Battisti, V., Redin, M., Linares, C.E.B., Dressler, V.L., Flores, M.M., Nicoloso, F.T., Morsch, V.M., Schetinger, M.R.C., 2006. Mercury toxicity induces oxidative stress in growing cucumber seedlings. Chemosphere 65 (6), 999–1106. https://doi.org/10.1016/j.chemosphere.2006.03.037.

Chen, J., Shiyab, S., Han, F.X.X., Monts, D.L., Waggoner, C.A., Yang, Z.M., Su, Y., 2009. Bioaccumulation and physiological effects of mercury in *Pterisvittata* and *Nephrolepis exaltata*. Ecotoxicology 18 (1), 110–121. https://doi.org/10.1007/s10646-008-0264-3.

Chinese National Standard Agency, 2017. Tolerance Limits of Contaminants in Foods. GB 2762-2017. pp. 1–17 (in Chinese).

Connell, J.H., 1979. Intermediate-disturbance hypothesis. Science 204, 1345. https://doi.org/10.1126/science.204.4399.1345.

Fay, L., Gustin, M., 2007. Assessing the influence of different atmospheric and soil mercury concentrations on foliar mercury concentrations in a controlled environment. Water Air Soil Pollut. 181, 373–384. https://doi.org/10.1007/s11270-006-9308-6.

Feng, X.B., Li, P., Qiu, G.L., Shang, L.H., Meng, B., Jiang, H.M., Bai, W.Y., Li, Z.G., Fu, X.W., 2008. Human exposure to methylmercury through rice intake in mercury mining areas, Guizhou province. China. Environ. Sci. Technol. 42 (1), 326–332. https://doi.org/10.1021/es071948x.

Fleming, E.J., Mack, E.E., Green, P.G., Nelson, D.C., 2006. Mercury methylation from unexpected sources: molybdate-inhibited freshwater sediments and an iron-reducing bacterium. Appl. Environ. Microbiol. 72 (1), 457–464. https://doi.org/10.1128/AEM.72.1.457-464.2006.

Frossard, A., Hartmann, M., Frey, B., 2017. Tolerance of the forest soil microbiome to increasing mercury concentrations. Soil Biol. Biochem. 105, 162–176. https://doi.org/10.1016/j.soilbio.2016.11.016.

Gerbig, C.A., Ryan, J.N., Aiken, G.R., 2012. The effects of dissolved organic matter on mercury biogeochemistry. In: Liu, G.L., Cai, Y., O'Driscoll, N. (Eds.), Environmental Chemistry and Toxicology of Mercury. John Wiley and Sons Inc., pp. 259–292 https://doi.org/10.1002/9781118146644.ch8.

Gilmour, C.C., Henry, E.A., Mitchell, R., 1992. Sulfate stimulation of mercury methylation in freshwater sediments. Environ. Sci. Technol. 26, 2281–2287. https://doi.org/10.1021/es00035a029.

Gilmour, C.C., Podar, M., Bullock, A.L., Graham, A.M., Brown, S.D., Somenahally, A.C., Johs, A., Hunt Jr., R.A., Bailey, K.L., Elias, D.A., 2013. Mercury methylation by novel microorganisms from new environments. Environ. Sci. Technol. 47 (20), 11810–11820. https://doi.org/10.1021/es403075t.

Haichar, F.E.Z., Heulin, T., Guyonnet, J.P., Achouak, W., 2016. Stable isotope probing of carbon flow in the plant holobiont. Curr. Opin. Biotechnol. 41, 9–13. https://doi.org/10.1016/j.copbio.2016.02.023.

Halliwell, B., Cutteridge, J.M.C., 1990. Role of free radicals and catalytic metal ions in human disease: an overview. Meth. Enzymol. 86, 1–85. https://doi.org/10.1016/0076-6879(90)86093-B.

Harris-Hellal, J., Vallaeys, T., Garnier-Zarli, E., Bousserrhine, N., 2009. Effects of mercury on soil microbial communities in tropical soils of French Guyana. Appl. Soil Ecol. 41 (1), 59–68. https://doi.org/10.1016/j.apsoil.2008.08.009.

Hernandez, C.E.M., Guerrero, I.E.P., Hernandez, G.A.G., Soils, E.S., Guzman, J.C.T., 2010. Catalase overexpression reduces the germination time and increases the pathogenicity of the fungus *Metarhizium anisopliae*. Appl. Microbiol. Biotechnol. 87 (3), 1033–1044. https://doi.org/10.1007/s00253-010-2517-3.

Horvat, M., Nolde, N., Fajon, V., Jereb, V., Logar, M., Lojen, S., Jacimovic, R., Falnoga, I., Qu, L.Y., Faganeli, J., Drobne, D., 2003. Total mercury, methylmercury and selenium in mercury polluted areas in the province Guizhou. China. Sci. Total Environ. 304, 231–256. https://doi.org/10.1016/S0048-9697(02)00572-7.

Hu, H., Li, M., Wang, G.X., Drosos, M., Li, Z., Hu, Z.Y., Xi, B.D., 2019. Water-soluble mercury induced by organic amendments affected microbial community assemblage in mercury-polluted paddy soil. Chemosphere 236, 124405. https://doi.org/10.1016/j.chemosphere.2019.124405.

Janero, D.R., 1900. Malondialdehyde and thiobarbituric acid-reactivity as diagnostic indices of lipid peroxidation and peroxidative tissue injury. Free Radic. Biol. Med. 9 (6), 515–540. https://doi.org/10.1016/0891-5849(90)90131-2.

Kainz, M., Lucotte, M., 2006. Mercury concentrations in lake sediments – revisiting the predictive power of catchment morphometry and organic matter composition. Water Air Soil Pollut. 170, 173–189. https://doi.org/10.1007/s11270-006-3009-z.

Kerin, E.J., Gilmour, C., Roden, E., Suzuki, M., Coates, J., Mason, R., 2006. Mercury methylation by dissimilatory iron-reducing bacteria. Appl. Environ. Microbiol. 72 (12), 7919–7921. https://doi.org/10.1128/AEM.01602-06.

Krabbenhoft, D.P., Sunderland, E.M., 2013. Global change and mercury. Science 341 (6153), 1457–1458. https://doi.org/10.1126/science.1242838.

Krupp, E.M., Mestrot, A., Wielgus, J., Meharg, A.A., Fledmann, J., 2009. The molecular form of mercury in biota: identification of novel mercury peptide complexes in plants. Chem. Commun. 28, 4257–4259. https://doi.org/10.1039/b823121d.

Li, M., Sherman, L.S., Blum, J.D., Grandjean, P., Mikkelsen, B., Weihe, P., Sunderland, E.M., Shine, J.P., 2014. Assessing sources of human methylmercury exposure using stable mercury isotopes. Environ. Sci. Technol. 48 (15), 8800–8806. https://doi.org/10.1021/es500340r.

Li, Y.Y., Zhao, J.T., Zhang, B.W., Liu, Y.J., Xu, X.H., Li, Y.F., Li, B., Gao, Y.X., Chai, Z.F., 2016. The influence of iron plaque on the absorption, translocation and transformation of mercury in rice (*Oryza sativa* L.) seedlings exposed to different mercury species. Plant Soil 398, 87–97. https://doi.org/10.1007/s11104-015-2627-x.

Lin, C.C., Yee, N., Barkay, T., 2012. Microbial transformations in the mercury cycle. In: Liu, G., Cai, Y., O'Driscoll, N. (Eds.), Environmental Chemistry and Toxicology of Mercury. John Wiley and Sons, Inc., pp. 155–191 https://doi.org/10.1002/9781118146644.ch5.

Liu, Y.T., Chen, Z.S., Hong, C.Y., 2011. Cadmium-induced physiological response and antioxidant enzyme changes in the novel cadmium accumulator, Tagetes patula. J. Hazard. Mater. 189 (3), 724–731. https://doi.org/10.1016/j.jhazmat.2011.03.032.

Liu, Y.R., Wang, J.J., Zheng, Y.M., Zhang, L.M., He, J.Z., 2014. Patterns of bacterial diversity along a long term mercury contaminated gradient in the paddy soils. Microbiol. Ecol. 68, 575–583. https://doi.org/10.1007/s00248-014-0430-5.

Liu, Y.R., Johs, A., Bi, L., Lu, X., Hu, H.W., Sun, D., He, J.Z., Gu, B.H., 2018. Unraveling microbial communities associated with methylmercury production in paddy soils. Environ. Sci. Technol. 52, 13110–13118. https://doi.org/10.1021/acs.est.8b03052.

Lu, G.W., Casaretto, J.A., Ying, S., Mahmood, K., Liu, F., Bi, Y.M., Rothstein, S.J., 2017. Overexpression of OsGATA12 regulates chlorophyll content, delays plant senescence and improves rice yield under high density planting. Plant Mol. Biol. 94, 215–227. https://doi.org/10.1007/s11103-017-0604-x.

Ma, M., Du, H.X., Wang, D.Y., Kang, S.C., Sun, T., 2017. Biotically mediated mercury methylation in the soils and sediments of Nam Co Lake, Tibetan Plateau. Environ. Pollut. 227, 243–251. https://doi.org/10.1016/j.envpol.2017.04.037.

Ma, M., Du, H.X., Sun, T., An, S.W., Yang, G., Wang, D.Y., 2019. Characteristics of archaea and bacteria in rice rhizosphere along a mercury gradient. Sci. Total Environ. 650, 1640–1651. https://doi.org/10.1016/j.scitotenv.2018.07.175.

Mcnutt, M., 2013. Mercury and health. Science 341 (6153), 1430. https://doi.org/10.1126/science.1245924.

Meng, B., Feng, X.B., Qiu, G.L., Cai, Y., Wang, D.Y., Li, P., Shang, L.H., Sommar, J., 2010. Distribution patterns of inorganic mercury and methylmercury in tissues of rice (*Oryza sativa* L.) plants and possible bioaccumulation pathways. J. Agric. Food Chem. 58, 4951–4958. https://doi.org/10.1021/jf904557x.

Meng, B., Feng, X.B., Qiu, G.L., Liang, P., Li, P., Chen, C.X., Shang, L.H., 2011. The process of methylmercury accumulation in rice (*Oryza sativa* L.). Environ. Sci. Technol. 45, 2711–2717. https://doi.org/10.1021/es103384v.

Meng, M., Li, B., Shao, J.J., Wang, T., He, B., Shi, J.B., Ye, Z.H., Jiang, J.B., 2014. Accumulation of total mercury and methylmercury in rice plants collected from different mining areas in China. Environ. Pollut. 184, 179–186. https://doi.org/10.1016/j.envpol.2013.08.030.

Messer, R.L., Lockwood, P.E., Tseng, W.Y., Edwards, K., Shaw, M., Caughman, G.B., Lewis, J.B., Wataha, J.C., 2005. Mercury (II) alters mitochondrial activity of monocytes at sublethal doses via oxidative stress mechanisms. J. Biomed. Mater. Res. Part B Appl. Biomater. 75 (2), 257–263. https://doi.org/10.1002/jbm.b.30263.

Nagajyoti, P.C., Lee, K.D., Sreekanth, T.V.M., 2010. Heavy metals, occurrence and toxicity for plants: a review. Environ. Chem. Lett. 8 (3), 199–216. https://doi.org/10.1007/s10311-010-0297-8.

Obrist, D., Kirk, J.L., Zhang, L., Sunderland, E.M., Jiskra, M., Selin, N.E., 2018. A review of global environmental mercury processes in response to human and natural perturbations: changes of emissions, climate, and land use. Ambio 47 (2), 116–140. https://doi.org/10.1007/s13280-017-1004-9.

Patra, M., Sharma, A., 2000. Mercury toxicity in plants. Bot. Rev. 66, 379–422. https://doi.org/10.1007/BF02868923.

Peng, S.B., Tang, Q.Y., Fang, F., Zou, Y.B., 2009. Current status and challenges of rice production in China. Plant Prod. Sci. 12 (1), 3–8. https://doi.org/10.1626/pps.12.3.

Philippot, L., Cregut, M., Che'neby, D., Bressan, M., Dequiet, S., Martin-Laurent, F., Ranjard, L., Lemanceau, P., 2008. Effect of primary mild stresses on resilience and resistance of the nitrate reducer community to a subsequent severe stress. FEMS Microbiol. Lett. 285 (1), 51–57. https://doi.org/10.1111/j.1574-6968.2008.01210.x.

Prasad, A.K.V.S.K., Saradhi, P.P., 1995. Effect of zinc on free radical and proline in *Brasica juncea* and *Cajanus cajan*. Phytochem. 39, 45–47. https://doi.org/10.1016/0031-9422(94)00919-K.

Prasad, K.V.S.K., Saradhi, P.P., Sharmila, P., 1999. Concerted action of antioxidant enzyme and curtailed growth under zinc toxicity in *Brassica juncea*. Environ. Exp. Bot. 42, 1–10. https://doi.org/10.1016/S0098-8472(99)00013-1.

Qiu, G.L., Feng, X.B., Wang, S.F., Shang, L.H., 2006. Environmental contamination of mercury from Hg-mining areas in Wuchuan, northeastern Guizhou, China. Environ. Pollut. 142 (3), 549–558. https://doi.org/10.1016/j.envpol.2005.10.015.

Qiu, G.L., Feng, X.B., Li, P., Wang, S.F., Li, G.H., Shang, L.H., Fu, X., Fu, X.W., 2008. Methylmercury accumulation in rice (*Oryza sativa* L.) grown at abandoned mercury mines in Guizhou, China. J. Agric. Food Chem. 56, 2465–2468. https://doi.org/10.1021/jf073391a.

Rieder, S.R., Frey, B., 2013. Methyl-mercury affects microbial activity and biomass, bacterial community structure but rarely the fungal community structure. Soil Biol. Biochem. 64, 164–173. https://doi.org/10.1016/j.soilbio.2013.04.017.

Rothenberg, S.E., Windham-Myers, L., Creswell, J.E., 2014. Rice methylmercury exposure and mitigation: a comprehensive review. Environ. Res. 133, 407–423. https://doi.org/10.1016/j.envres.2014.03.001.

Selin, N.E., 2018. A proposed global metric to aid mercury pollution policy. Science 360 (6389), 607–609. https://doi.org/10.1126/science.aar8256.

Shah, K., Dubey, R.S., 1998. Effect of cadmium on proline accumulation and ribonuclease activity in rice seedlings: role of proline as a possible enzyme protectant. Biol. Plant. 40, 121–130. https://doi.org/10.1023/A:1000956803911.

Strickman, R.J., Mitchell, C.P., 2017. Accumulation and translocation of methylmercury and inorganic mercury in Oryza sativa: an enriched isotope tracer study. Sci. Total Environ. 574, 1415–1423. https://doi.org/10.1016/j.scitotenv.2016.08.068.

Sun, S.W., Lin, Y.C., Weng, Y.M., Chen, M.J., 2006. Efficiency improvements on ninhydrin method for amino acid quantification. J. Food Compost. Anal. 19, 112–117. https://doi.org/10.1016/j.jfca.2005.04.006.

Tang, Z.Y., Fan, F.L., Wang, X.Y., Shi, X.J., Deng, S.P., Wang, D.Y., 2017. Mercury in rice (*Oryza sativa* L.) and rice-paddy soils under long-term fertilizer and organic amendment. Ecotoxicol. Environ. Saf. 150, 116–121. https://doi.org/10.1016/j.ecoenv.2017.12.021.

Ullrich, S.M., Tanton, T.W., Abdrashitova, S.A., 2001. Mercury in the aquatic environment: a review of factors affecting methylation. Crit. Rev. Environ. Sci. Technol. 31 (3), 241–293. https://doi.org/10.1080/20016491089226.

Venturi, V., Keel, C., 2016. Signaling in the rhizosphere. Trends Plant Sci. 21 (3), 187–198. https://doi.org/10.1016/j.tplants.2016.01.005.

Verma, S., Dubey, R.S., 2003. Lead toxicity induces lipid peroxidation and alters the activities of antioxidant enzymes in growing rice plants. Plant Sci. 164 (4), 645–655. https://doi.org/10.1016/S0168-9452(03)00022-0.

Wyn, B., Kidd, K.A., Burgess, N.M., Curry, R.A., 2009. Mercury biomagnification in the food webs of acidic lakes in Kejimkujik National Park and National Historic Site. Nova Scotia. Can. J. Fish. Aquat. Sci. 66 (9), 1532–1545. https://doi.org/10.1139/F09-097.

Xu, X.H., Lin, Y., Meng, B., Feng, X.B., Xu, Z.D., Jiang, Y.P., Zhong, W.L., Hu, Y.H., Qiu, G.L., 2017. The impact of an abandoned mercury mine on the environment in the Xiushan region, Chongqing, southwestern China. Appl. Geochem. 88, 267–275. https://doi.org/10.1016/j.apgeochem.2017.04.005.

Xu, X.H., Yan, M., Liang, L.C., Han, J.L., Liu, L., Feng, X.B., Guo, J.Y., Wang, Y.J., Qiu, G.L., 2019. Impacts of selenium supplementation on soil mercury speciation, and inorganic mercury and methylmercury uptake in rice (*Oryza sativa* L.). Environ. Pollut. 249, 647–654. https://doi.org/10.1016/j.envpol.2019.03.095.

Zhang, W.H., Tyerman, S.D., 1999. Inhibition of water channels by $HgCl_2$ in intact wheat root cells. Plant Physiol. 120 (3), 849–857. https://doi.org/10.1104/pp120.3.849.

Zhang, H., Feng, X.B., Larssen, T., Qiu, G.L., Vogt, R.D., 2010a. In inland China, rice, rather than fish, is the major pathway for methylmercury exposure. Environ. Health Perspect. 118 (9), 1183–1188. https://doi.org/10.1289/ehp.1001915.

Zhang, H., Feng, X.B., Larssen, T., Shang, L.H., Li, P., 2010b. Bioaccumulation of methylmercury versus inorganic mercury in rice (*Oryza sativa* L.) grain. Environ. Sci. Technol. 44 (12), 4499–4504. https://doi.org/10.1021/es903565t.

Zhou, Z.S., Huang, S.Q., Guo, K., Mehta, S.K., Zhang, P.C., Yang, Z.M., 2007. Metabolic adaptations to mercury-induced oxidative stress in roots of Medicago sativa L. J. Inorg. Biochem. 101, 1–9. https://doi.org/10.1016/j.jinorgbio.2006.05.011.